UNIVERSITY OF CALGARY
FACULTY OF SCIENCE
Department of Physics and Astronomy

Hydrogen Balmer Series

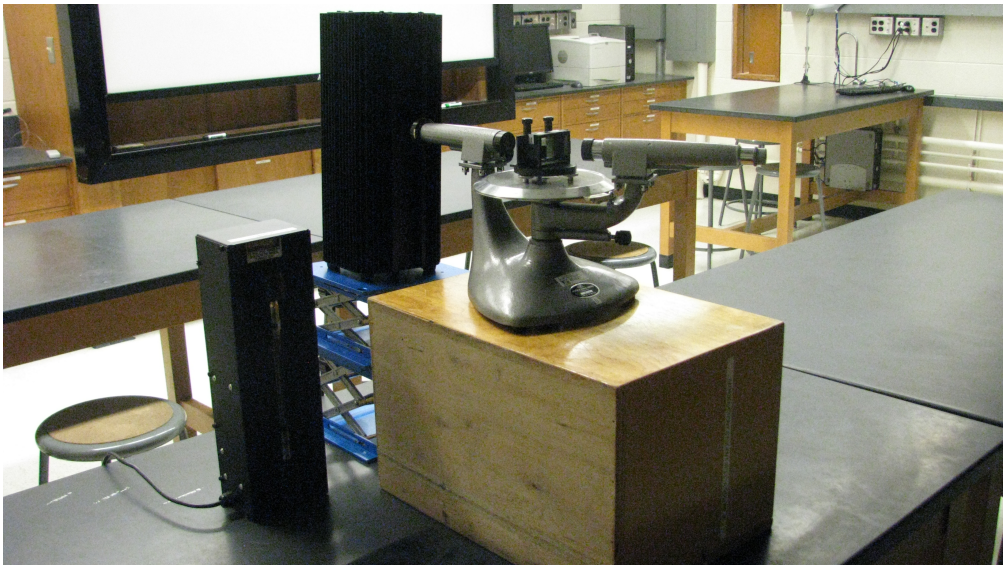


Figure 1.1. Hydrogen Balmer series experimental setup.

Equipment: Introductory student spectroscope, 600 lines/mm diffraction grating, Geissler tube power supply, hydrogen Geissler tube, laboratory jack [2], magnifying glass

Goals of the Experiment: To learn how to calibrate a spectroscope using a spectral source with known wavelengths. To measure angles of diffraction and from these, determine experimental values for the wavelength of each spectral line in the hydrogen Balmer series. To determine an experimental values for the diffraction grating line spacing and Rydberg constant.

Theory

The Hydrogen Atom

With the simplest spectrum, hydrogen is the most basic case of emission and absorption to study. Excited atoms can emit electromagnetic radiation at many wavelengths but for the purposes of this experiment, only the visible region of the electromagnetic spectrum is examined. The visible emission spectrum of hydrogen is known as the **Balmer Series**

Empirical **Balmer formula** takes the form:

$$\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{n^2}\right], \quad n = 3, 4, 5, ... \tag{1.1}$$

where R is the **Rydberg's constant** and each integer, n, corresponds to an observed hydrogen spectrum line.

In 1913, Niels Bohr's theory for atomic structure was able to account for the observed behaviour of atoms, including Balmer's formula. The problem that Bohr addressed was the motion of a single electron about a nucleus of charge Z, where Z is the **atomic number**, the number of protons in the nucleus. Following Bohr, assume that the electron moves with velocity v in an orbit about the nucleus and when an electron remains in a **stable orbit** it does not radiate energy. Furthermore, assume that mass of the nucleus, m_Z, is much larger than mass of the electron, m_e, so the electron's orbit is a circle with radius r. Under these conditions the Coulomb force is the source of the centripetal acceleration of the orbiting electron, which gives the expression

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2 Z}{r^2} = \frac{m_e v^2}{r} \tag{1.2}$$

where $\varepsilon_0 = 8.854 \times 10^{-12}$ C^2 N^{-1} m^{-2} is the permittivity of free space, $e = 1.602 \times 10^{-19}$ C is the electronic charge and $m_e = 9.109 \times 10^{-31}$ kg is the mass of the electron

The momentum condition for stable electron orbits is given by the expression

$$2\pi m_e v r = nh, \quad n = 1, 2, 3, ... \tag{1.3}$$

where $h = 6.626 \times 10^{-34}$ m^2 kg s^{-1} is Planck's constant and n is the principle quantum number. Only orbits designated by n, are stable.

The total energy of the electron, E, is

$$E = K + U = -\frac{1}{8\pi\varepsilon_0}\frac{e^2 Z}{r} = -\frac{1}{n^2}\,\frac{m_e e^4 Z^2}{8\varepsilon_0^2 h^2}, \tag{1.4}$$

where K is the kinetic energy of the electron and U is electrostatic potential energy of the system.

Each electron orbit must have a quantized amount of energy specified by the quantum number n. Note that the energy function is negative, so as n increases, the energy becomes less negative, from the lowest energy level for $n = 1$ to maximum of zero at $n = \infty$.

An electron only emits radiation when it changes orbits and there is a change in energy ΔE, for this transition. The energy of the emitted photon is given by the Planck formula

$$\Delta E = hf = \frac{hc}{\lambda} \tag{1.5}$$

where $c = 2.998 \times 10^8$ m s^{-1} is the speed of light, f is the **frequency** of the photon, and λ is the **wavelength** of the photon. Notice that the frequency of the emitted photon is unrelated to the frequency with which the electron circles the nucleus.

Let the quantum number of the initial orbit be n_i and the quantum number of the final orbit be n_f. Substituting the initial and final energies given by Equation 1.4 into Equation 1.5 leads to

$$\frac{1}{\lambda} = \frac{m_e e^4 Z^2}{8\varepsilon_0^2 h^3 c}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right). \tag{1.6}$$

Comparing Equation 1.6 with Equation 1.1 indicates that the Rydberg constant is

$$R = \frac{m_e e^4 Z^2}{8\varepsilon_0^2 h^3 c}. \tag{1.7}$$

These results provide an explanation for the spectral lines of the Balmer series When an atom is excited, the increase in energy is accounted for when the electron jumps to a higher energy level. Emission lines result when the electron jumps back down to a lower energy level and emits a photon. In the case of hydrogen, where $Z = 1$, the visible Balmer lines arise when an electron jumps down to the $n_f = 2$ orbit. A comparison of the experimentally obtained Rydberg constant with the constant predicted by Equation 1.7 tests Bohr's atomic theory. As there is only a specific number of energy levels, the photons created have correspondingly specific wavelengths giving rise to a discrete spectrum.

Figure 1.2 depicts some of the electron transitions in the hydrogen atom.

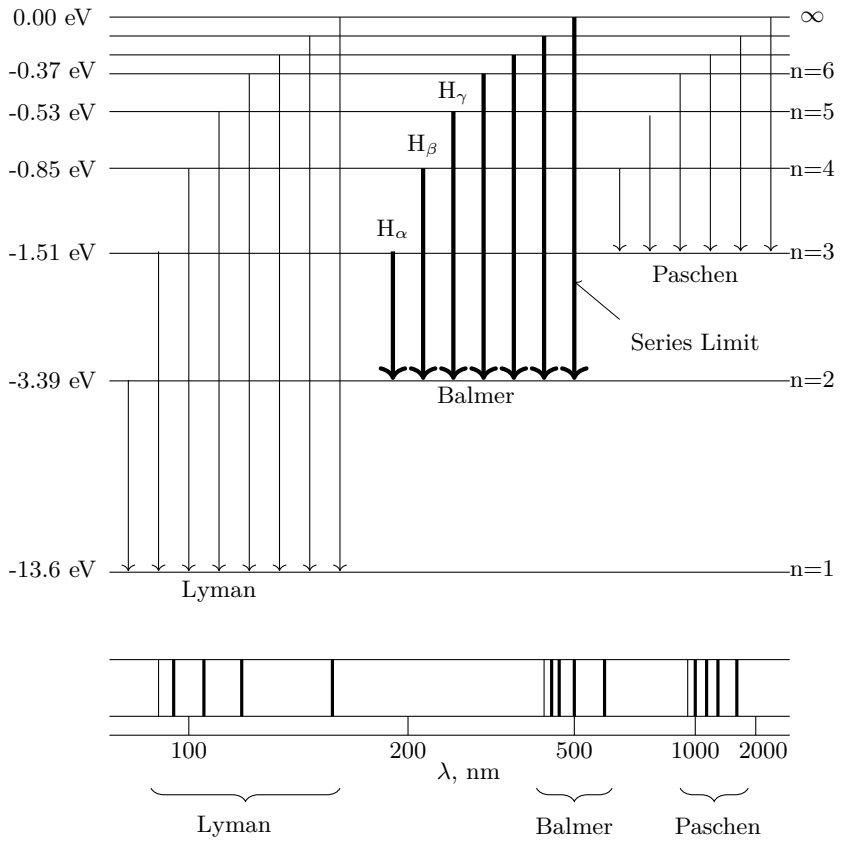


Figure 1.2. Electronic de-excitation in hydrogen with corresponding spectral lines. The figure includes transitions in the ultraviolet (**Lyman series**, $n_f = 1$), the visible (**Balmer series**, $n_f = 2$), and the infrared regions (**Paschen series**, $n_f = 3$) of the spectrum. The energy of each level is indicated on the left side and the respective quantum numbers on the right, as determined by Equation 1.4

The Spectroscope

The spectral lines visible to the human eye vary from violet to red in colour. The most common method of observing spectra is through the use of a **spectroscope**. Figure 1.3 is a diagram of the Spectroscope used in this experiment. Light passes through the **slit** and the **collimator**. It is then bent by the **diffraction grating**. The **Telescope** is connected to a rotating arm so that it can be moved to allow the bent light to pass through it. A vernier on the telescope is then used to measure the angle at which the incident light is diffracted. This angle can be used to find the wavelength of the selected emission line. Encircling the base of the grating table is a 360 degree scale used to make measurements in conjunction with the vernier scale fixed to the telescope arm.

Before the spectroscope can be used to measure the angles of spectral lines it must be adjusted to ensure that accurate results are obtained. To begin, the telescope is focused on a distant object, using the **telescope focus**. The telescope is then lined up with the slit in the end of the collimator, and the **collimator focus** is adjusted until the slit comes into sharp focus. In this way only parallel rays of light pass through the **diffraction grating**.

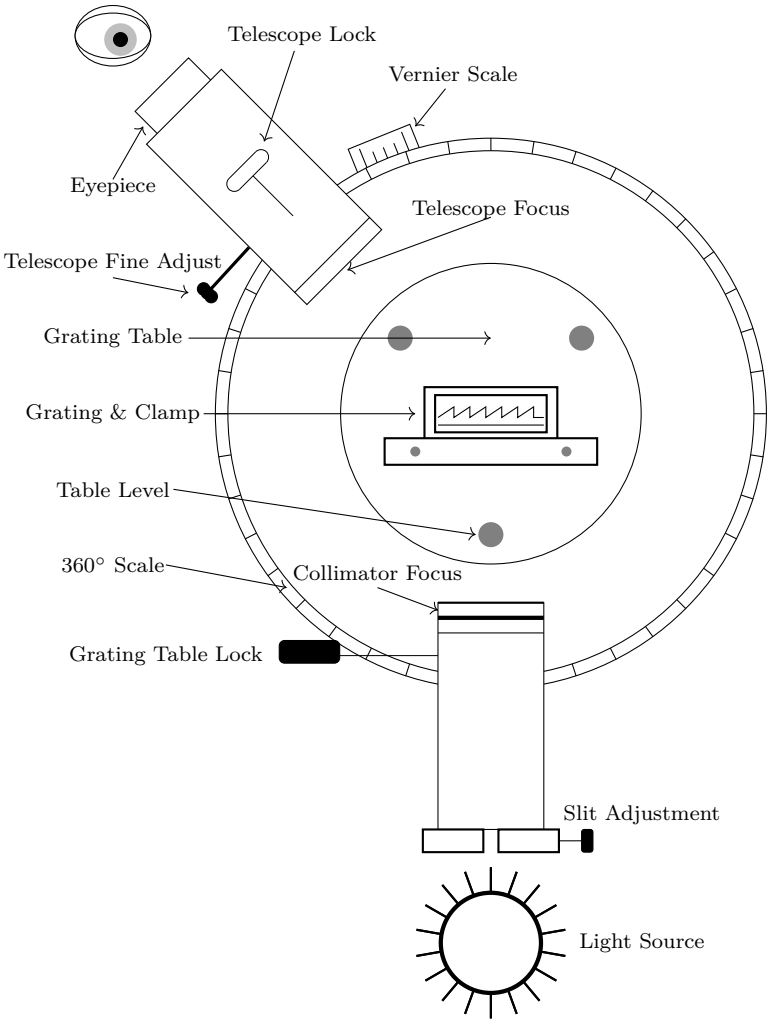


Figure 1.3. A diagram of the spectrometer used in this experiment.

Diffraction Gratings

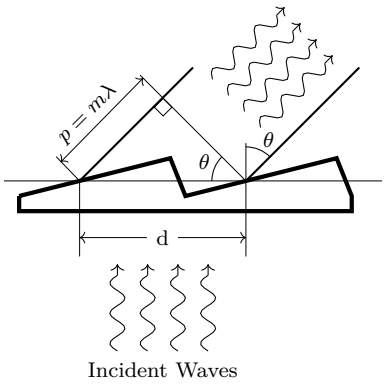


Figure 1.4. Diffraction of incident light by grating.

A schematic picture of a diffraction grating is shown in Figure 1.4. When light from an emission spectrum passes through a diffraction grating the light is split into its constituent colours. The condition for constructive

interference is

$$d \sin\theta = m\lambda \qquad m = 0, \pm 1, \pm 2, ... \tag{1.8}$$

where d is the **line spacing**, that is, the distance from one point on a groove to the same point on the neighboring groove. Line spacing is an inverse of line density of the diffraction grating.

The **angle of deviation**, θ, is the angle between the central maximum and the spectral line. The variable, m, indicates the **order** of the spectral lines. First order spectral lines ($m = \pm 1$) are those found closest to the central maximum. As the telescope arm moves away from the central maximum, the higher order spectral lines ($m = \pm 2, \pm 3, ...$) can be observed. Only first order maxima are used in this experiment.

Spectral Sources

Helium	
Wavelength (nm)	Colour
667.8150	bright red
587.5620	bright yellow
501.5678	bright green
471.3146	bright blue
447.1479	bright purple

Table 1.1. Spectral line wavelengths and apparent colours for helium.

The light from the emission spectrum that passes through the diffraction grating is generated by lamps called **spectral sources**. Three different types of spectral sources used in this experiment are: the hydrogen and helium Geissler tubes, the low pressure sodium lamp, and the Osram cadmium lamp.

In this experiment the spectral lines of helium, sodium or cadmium are used to calibrate the spectroscope and determine an experimental value for the diffraction grating spacing. The accepted wavelengths for the primary visible spectral lines of these elements are presented in Table 1.1. Once calibration is complete, the Balmer series emission spectrum of hydrogen will be measured.

Safety Considerations & Technical Remarks

- Geissler spectral lamps should be turned off when not being used to lengthen their life. Be careful when handling a high voltage supply, do not touch any exposed metal parts.
- Do not adjust settings on the Ealing power supply if given a lamp with an external supply.

The Experiment

Uncertainty Estimation and Propagation

Usually $u(\theta)$ would be half the smallest division of the vernier on spectroscope, however there are other factors affect to the error. Those things include, but are not limited to the position of the eye, thickness of the spectral lines and crosshairs, limits on the resolution of the diffraction grating and the spectroscope. For these reasons, it is reasonable to estimate the uncertainty in measurements of angles for this spectroscope to be $0.2°$, converted to radians for the purpose of the calculations. The uncertainty in the line density of 600 lines/mm for the diffraction grating to be 5 lines/mm. All other uncertainties should be propagated using appropriate methods.

Pre-Lab

Your Pre-lab exercises are available on the D2L Lab site (Assessment – Quizzes) and should be completed there.

Experimental Procedure

$\Longrightarrow$ **Step 1** Remove the diffraction grating and its holder from the spectroscope. It is extremely important not to touch or clean **any** of the lenses or the diffraction grating.

$\Longrightarrow$ **Step 2** The telescope must be focused for parallel light rays and the crosshairs must be put into focus. First slide the eyepiece on the telescope in or out and bring the crosshairs into sharp focus. Next, focus the telescope on a distant object across the room.

$\Longrightarrow$ **Step 3** Use the helium lamp as the calibration light source and place it so that the light from it passes through the slit in the collimator. Aim the telescope at the slit and gently move the lamp until the slit is at its brightest. Adjust the collimator focus until the slit is in sharp focus. In this way the spectrum being measured uses only light parallel to the slit. Align the cross-hairs in the eyepiece with the slit. Do not change the focus of the telescope or the collimator, or rotate the eyepiece for the remainder of the experiment, as this affects the apparent position of the centre slit. Do not move the telescope by touching the eyepiece or the chrome part of the telescope, as this could change the focus of telescope.

$\Longrightarrow$ **Step 4** Replace the diffraction grating into its holder. The bolts for the diffraction grating holder should be facing the collimator, and the diffraction grating frame should be placed in the holder so that the diffraction grating is as close to the holder as possible. The idea here is that the diffraction grating needs to pass through the centre of the grating table as seen in Figure 1.3.

$\Longrightarrow$ **Step 5** Turn the telescope and look at spectral lines that are far away from the center angle. They should be vertically in the middle of the telescope as shown in Figure 1.5. If they are not leveled, adjust the level of the table until the lines are vertically in the middle of the telescope using the three leveling screws on the bottom of the grating table. Move the telescope to the other side of centre and adjust the lines until they are also centered. This adjustment may

have be repeated several times on each side.

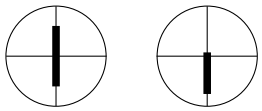


Figure 1.5. Levelling the grating table.

⟹ **Step 6** Look through the telescope and adjust the width of the slit. Take note of which side of the slit moves. All measurements should be taken from the side of the light line that **does not** move. By doing this the width of the slit can be changed during the experiment. The slit width adjustment is used to adjust the amount of light that is allowed to pass into the collimator, as different amounts of light are useful in different situations. Some lines are close together and will overlap if the slit is too wide, in which case a narrow slit is required in order to resolve all of the lines that are present. On the other hand some lines are very dim. In such a case a wide slit is better in order for the spectral line to be bright enough that it can be seen.

⟹ **Step 7** Align the diffraction grating so it is perpendicular to the collimator as seen in Figure 1.4. This is done by rotating the grating table. If the diffraction grating is not perpendicular, the angle of deviation of the spectral lines will differ between the two sides of the centre maximum. Choose any line in the spectrum and rotate the grating until the angle of deviation is the same on the left spectrum as the right spectrum. This step may need to be repeated several times in order to obtain satisfactory results.

⟹ **Step 8** Measure the angle of the centre maximum. Align the cross-hairs with the stationary edge of the slit. Use the vernier to measure the angle of the maximum. A magnifying glass has been provided to assist in reading the vernier. All angles of diffraction are measured with respect to this centre angle. Each angle of diffraction is subtracted from the centre angle and the absolute value of that angle is the angle of deviation. The angle of deviation is then used to determine wavelengths of spectral lines using Equation 1.8.

⟹ **Step 9** Use the helium spectral lamp to check the performance of the diffraction grating. The measured angle of deviation, along with Equation 1.8, can be used to calibrate the spectroscope. If Equation 1.8 holds, a plot of λ versus $\sin(\theta)$ should yield a straight line with slope d and an intercept of zero. Using the given wavelengths from Table 1.1 and measured angles of deviation, make a quick plot of all $\lambda_{\text{theoretical}}$ versus $\sin(\theta)$ (uncertainties need not be included here). Use the slope to determine the line spacing for the diffraction grating. Use uncertainty in the slope to determine the uncertainty of the line spacing d.

⟹ **Step 10** The spectroscope is now adjusted, checked, and ready for analyzing the hydrogen Balmer series. Turn on the hydrogen lamp and shine the light through the slit in the collimator. Measure the angle of deviation, on one side of the center maximum, for the four highest wavelengths calculated in the Pre-lab.

Lab Report

Your lab report should include the following items:

- Data
  - notes and observations from the setup and alignment of the spectroscope;
  - table(s) of data from the performance check of the diffraction grating; include calibration source, center angle, colour, angle of deviation, and theoretical wavelengths with necessary uncertainties;
  - table(s) of data from the hydrogen lamp
- Analysis:
  - a graph of $\lambda_{\text{theoretical}}$ versus $\sin(\theta)$ including the line of the best fit, fitting coefficients and their uncertainties;
  - a value of the line spacing d for the diffraction grating with uncertainty and all necessary calculations;
  - calculated experimental values of the wavelengths of the Balmer series with uncertainties;
  - a graph of λ^{-1} as a function of n^{-2}, including the line of the best fit, fitting coefficients and their uncertainties;
  - calculations of Rydberg's constant;
  - calculation of the ionization energy;
- Discussion:
  - a comparison of the measured and expected values;
  - comment on agreements/discrepancies between the values;
  - comment of shapes and fits of graphs;
  - possible sources of uncertainties; classification of errors as random/systematic.
- Conclusions